

BrickAtlas: Supplementary Evidence

Development Manuscript
Author metadata intentionally omitted

1. Benchmark Reference

This supplement separates three types of evidence: measured full-casebank audits, measured adaptive model studies, and explicitly unmeasured future experiments. The latter have blank result cells and must not be interpreted as zero accuracy. All new figures are generated from archived renders or machine-readable evidence, not illustrative model outputs.

1.1. Curated Calibration Cases

The 5,120-source v2 population remains unchanged and preserves all historical scores. A new `curated-casebank-1` layer provides twelve original, explicitly authored cases for qualitative inspection, task calibration, and future human studies. It is not counted as 5,132 v2 sources and is not a hidden confirmation set.

Every structure passes both geometry implementations and a piece-by-piece vertical insertion check. The examples span furniture (bench, picnic table), architecture (staircase, gate, pavilion, townhouse, station), infrastructure (single- and double-span bridges), and landmarks (watchtower, lighthouse, two-tier pagoda). They use two to four role-based colors rather than independently sampled part colors. “Hard” may denote a long dependency chain or multiple separated supports rather than only more pieces.

Each model directory includes `structure.json`, `metadata.json`, four views, every occupied layer, an exploded view, and legal assembly checkpoints. Each task directory includes the public task contract, expected output, input renders where applicable, and ground-truth renders. `CONTACT_SHEET.png`, `TASK_GROUND_TRUTH.png`, and `index.html` provide direct review entry points. The designs are original CC0 BrickAtlas assets; no official-set geometry is copied. Their semantic recognizability remains a claim to validate with independent people.

1.2. Advanced Challenge Layer

The measured v2 study exposes two opposite regimes: one-clause recoloring and removal are saturated, while ordinary reconstruction, completion, and shifted-pose repair often remain at zero exact success. Moreover, a height-sorted algorithm solves all v2 planning cases. The separate

`challenge-casebank-1` layer therefore changes both scale and task structure without modifying v2 or historical scores.

The contracts are recovery planning from a legal dead end, iterative support counterfactuals, a three-clause compound edit, simultaneous color/pose/extra-part repair, costed active inspection, shortest-path and articulation reasoning, non-suffix distributed completion, rigid subassembly translation, legal-next-part set prediction, selected-part pose estimation, 53-part reconstruction, and inventory-constrained redesign. Recovery actions are executed in submitted order. The counterfactual repeatedly removes newly unsupported parts. The selection oracle returns all legal candidates, not one arbitrary next piece. Multi-fault repair requires both exact restoration and the complete set of wrong current IDs.

This layer remains far smaller than BrickNet and lacks LDRAW connector diversity, force stability, manuals, and robot actions. Its purpose is controlled cross-capability stress testing, not a claim of comparable asset scale or generative realism.

1.3. Source Policies and Catalog

Figure 5 shows all eight construction procedures. Table 4 gives their measured means; height uses plate units, and multi-supported counts pieces with at least two distinct lower supporting parts. Each row contains 640 source objects. These differences are useful geometric controls but not eight independently collected semantic categories. The shared root plate makes retrieval and prior-shape controls especially important.

1.4. Construction and Expansion

For each policy, candidate seeds are deterministic. The pipeline is:

1. Generate a valid insertion sequence from the common root plate under the policy-specific attachment rule.
2. Normalize horizontal origin; reject invalid sizes and exact colorless translation/yaw-equivalent duplicates, including old releases.
3. Run the independent integer-cell geometry check. Accept until 640 unique geometries are available per policy.

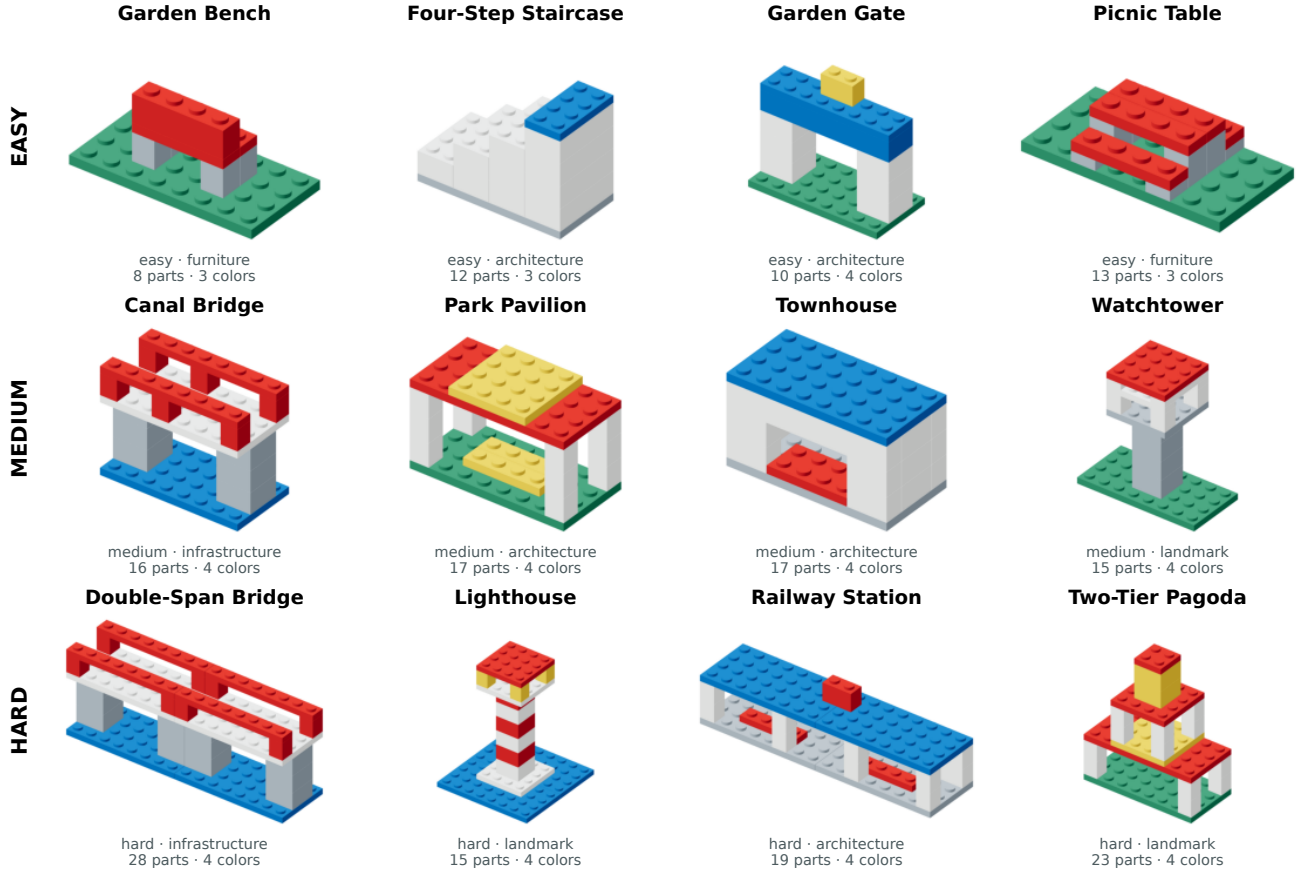


Figure 1. The complete curated set: four easy, four medium, and four hard structures. Difficulty combines part count, occupied levels, occlusion, support structure, and span length. Colors encode functional roles and never exceed four per object.

Model	Difficulty	Style	Parts	Colors	Levels	Occlusion / support
Garden Bench	easy	furniture	8	3	4	low / single
Four-Step Staircase	easy	architecture	12	3	6	low / single
Garden Gate	easy	architecture	10	4	7	low / multiple
Picnic Table	easy	furniture	13	3	4	low / multiple
Canal Bridge	medium	infrastructure	16	4	7	medium / spanning
Park Pavilion	medium	architecture	17	4	6	medium / multiple
Townhouse	medium	architecture	17	4	5	high / multiple
Watchtower	medium	landmark	15	4	10	medium / multiple
Double-Span Bridge	hard	infrastructure	28	4	7	medium / spanning
Lighthouse	hard	landmark	15	4	11	medium / multiple
Railway Station	hard	architecture	19	4	5	high / multiple
Two-Tier Pagoda	hard	landmark	23	4	11	high / multiple

Table 1. Curated-case specification. These labels describe explicit design contracts, not measured human difficulty. Exact structures, palettes, legal orders, views, layers, and steps are stored under `benchmark/curated-cases/`.

- Assign the source split from the geometry hash, except for the two entirely held-out policies.
- Derive task variants and reference conditions; attach the source group to every derivative; build the public input and evaluator record separately.
- Validate oracle success, empty-output failure, metric applicability, render recipes, and hashes before materializing compressed shards.

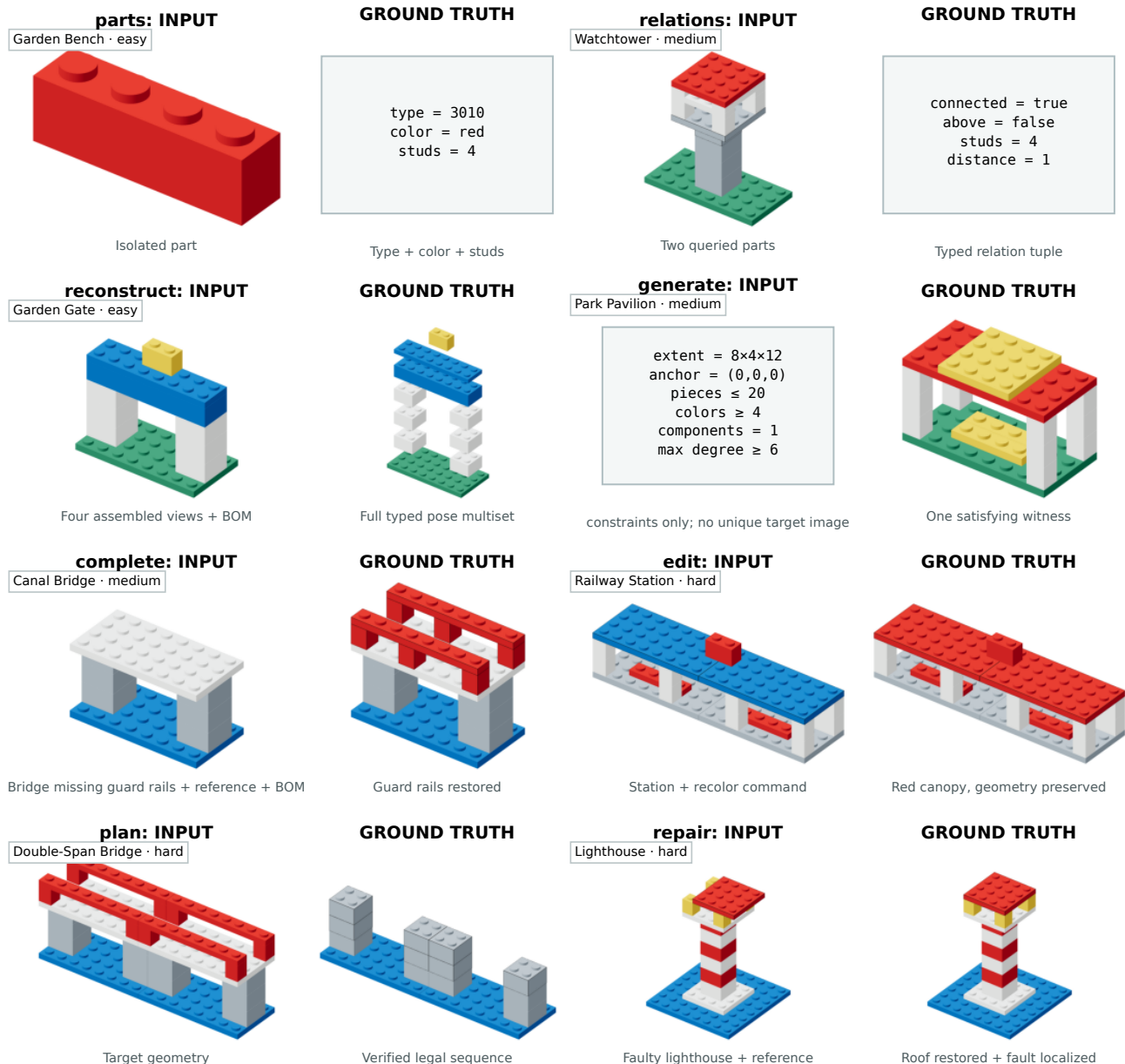


Figure 2. All eight curated task demonstrations. Text panels show non-image labels or constraints; visual panels show actual input structures and evaluator ground truth. Generation has no unique target, so the pavilion is explicitly one satisfying witness. Planning ground truth is a verified prefix, not merely a final object render.

The release contains 14 shards and 476,760 serialized rows across source, input, ground-truth, render, and case-index records. Those storage rows are not extra samples.

1.5. Input and Answer Schemas

The public exporter emits `caseId`, a fixed system prompt, the public input task, and relative images paths. The task contains a question, public structure/constraints where applicable, and a response schema. The full evaluator

index is public for reproducibility; ordinary-input isolation is a benchmark protocol, not a security guarantee against consulting the answers.

The following is a minimal *schema example*, not a model result:

```
{"version":1,"parts":[
  {"id":"p0","partId":"3005",
   "color":"blue","x":0,"y":0,
   "z":0,"turn":0}
```

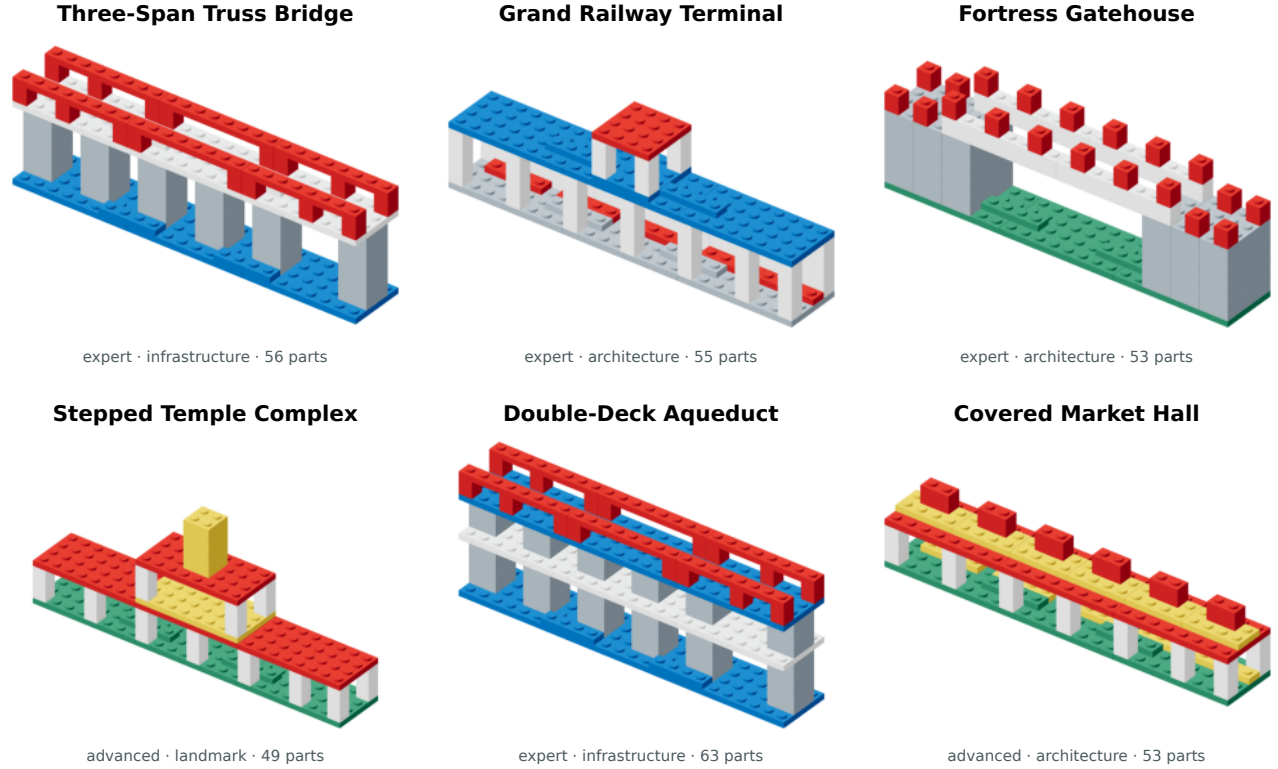


Figure 3. Six original challenge structures. They contain 49–63 parts each (329 placed instances total), compared with 8–28 parts in the curated calibration layer. All are connected and have verified piece-wise vertical insertion orders.

Model	Tier	Domain	Parts	Intended pressure
Three-Span Truss Bridge	expert	infrastructure	56	56-part long-horizon plan; Multiple legal frontiers; Recoverable deck-first dead end
Grand Railway Terminal	expert	architecture	55	55-part scene reconstruction; Repeated-instance disambiguation; Compound regional edit
Fortress Gatehouse	expert	architecture	53	53-part occlusion; Long graph paths; Inspection choice under hidden structure
Stepped Temple Complex	advanced	landmark	49	49-part distributed completion; Nested subassembly transform; Strong symmetry traps
Double-Deck Aqueduct	expert	infrastructure	63	Fine-grained next-part selection; Pose estimation among repeated bays; 63-part output length
Covered Market Hall	advanced	architecture	53	53-part multi-view reconstruction; Inventory-constrained redesign; Repeated-part correspondence

Table 2. Challenge structures and design intent. Tiers are controlled design strata, not human-calibrated difficulty labels.

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Recognition returns `partId`, `color`, `studs`. Relations return `connected`, `above`, `contactStuds`, `shortestPath`. Plans return `order`, an array of all target IDs exactly once. Repair wraps a full program as `structure` and adds `faultIds`; no-fault answers use an empty array. Normal reconstruction, generation, completion, and editing return a `structure`.

Relative nodes may instead specify an earlier parent, an offset in global grid axes, and absolute turn. This encoding is reversible but does not assert a stud connection; it is not BrickNet’s typed connector language. Square footprints admit all quarter-turns, while other rectangular footprints admit half-turn symmetry. Scoring ignores instance IDs for multiset recovery but does not globally align translations or rotations unless the task explicitly requests a transformation.

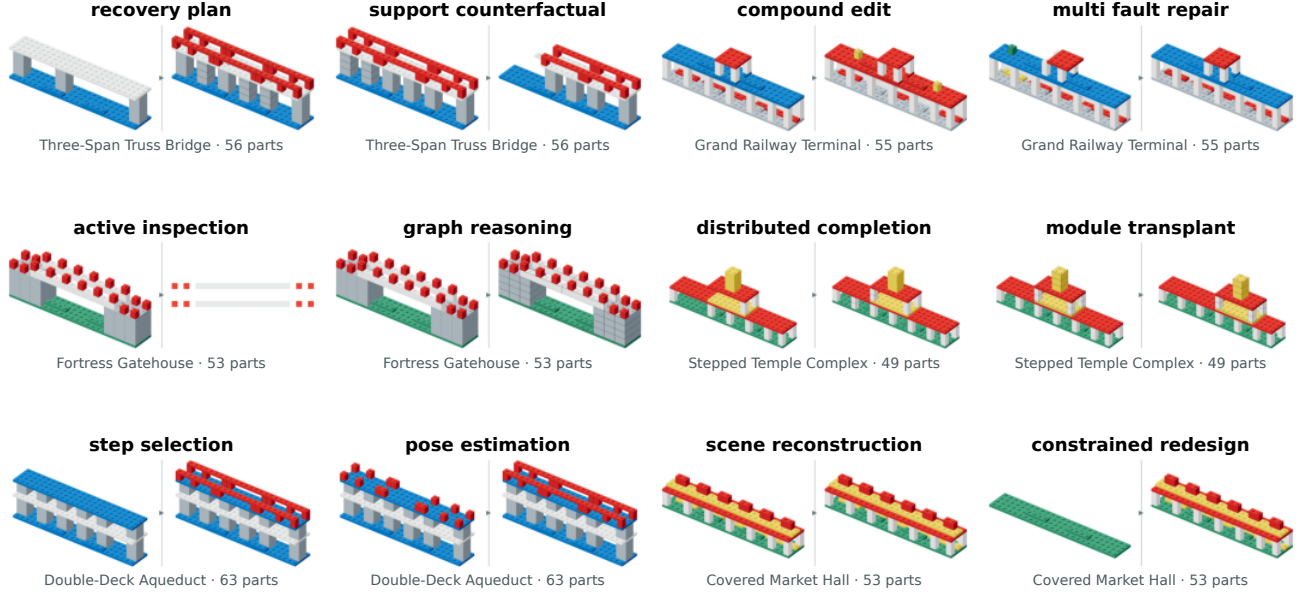


Figure 4. Twelve advanced contracts, each shown as input/context (left) and oracle target or disclosed consequence (right). Visual targets do not replace the typed JSON oracle. Constrained redesign accepts any satisfying structure rather than only the displayed witness.

Task	Source	Parts	Model result
recovery-plan	Three-Span Truss Bridge	56	
support-counterfactual	Three-Span Truss Bridge	56	
compound-edit	Grand Railway Terminal	55	
multi-fault-repair	Grand Railway Terminal	55	
active-inspection	Fortress Gatehouse	53	
graph-reasoning	Fortress Gatehouse	53	
distributed-completion	Stepped Temple Complex	49	
module-transplant	Stepped Temple Complex	49	
step-selection	Double-Deck Aqueduct	63	
pose-estimation	Double-Deck Aqueduct	63	
scene-reconstruction	Covered Market Hall	53	
constrained-redesign	Covered Market Hall	53	

Table 3. **Unmeasured.** The challenge task list. Blank cells are deliberate: oracle and near-miss validation are complete, but no model has been run on these new cases.

1.6. Render and Coordinate Contract

The released views are 640-by-480 PNGs. X , Z are stud coordinates; Y is bottom plate height. The front camera looks from positive Z , with image-right $+X$ and image-up $+Y$. The side camera looks from positive X , with image-right $-Z$ and image-up $+Y$. Top looks from positive Y with a small positive- Z offset, image-right $+X$ and image-down $+Z$. Cameras fit each structure; apparent pixel scale is not a fixed metric calibration. Layer views remove every part not starting at the selected bottom Y . They are additional observations, not ordinary views through an opaque object.

1.7. Full No-Operation Audit

`study/paper-audit.ts` first verifies frozen casebank source hashes and the compressed structure shard digest, computes a per-source inventory, and evaluates the copy predictor on 35,840 selected cases. The predictor receives only the public task and copies `current`; repair adds an empty fault list. Prediction production does not inspect the evaluator target, changed IDs, or oracle. Its fixed output is then scored with the versioned strict evaluator.

Seven endpoints each contain one case per source: ordinary suffix completion, three default symbolic edits, and ordinary no-fault/color/shift repair. Raw case-level scores, IDs, policies, and splits are retained in

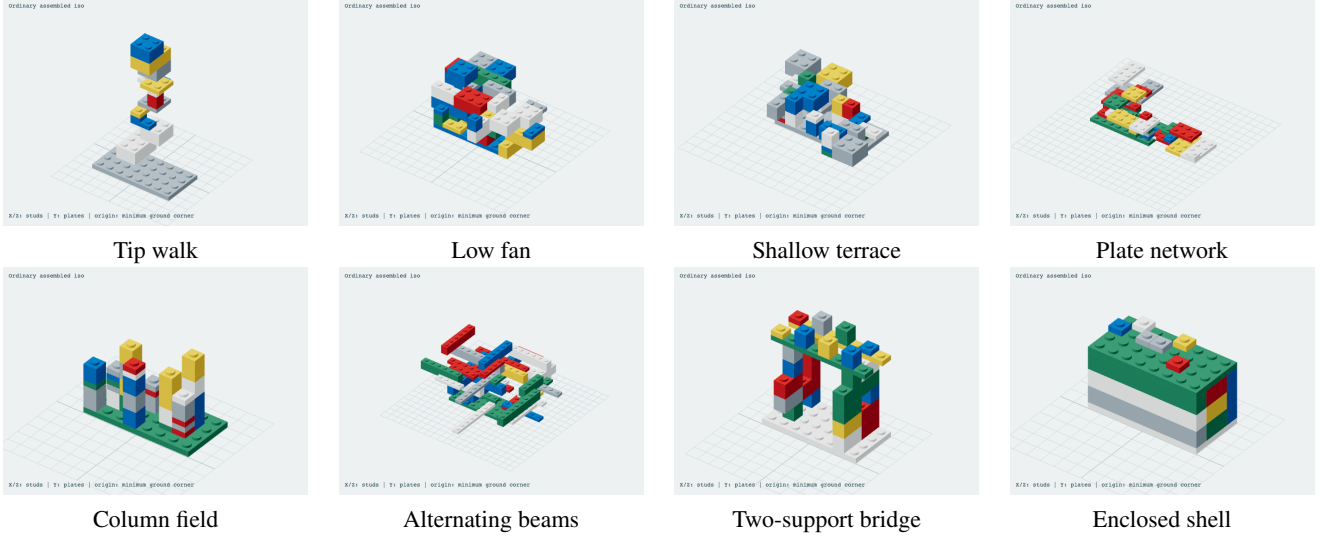


Figure 5. Actual largest-case-per-policy render-audit examples. Bridge and shell are policy-held-out. Selection stresses rendering complexity rather than providing a random visual summary.

Policy	N	Parts	Height	Multi-sup.
tip-walk	640	11.3	25.7	0.00
low-fan	640	25.0	12.4	0.75
shallow-terrace	640	25.0	6.9	0.30
plate-network	640	25.0	8.3	1.64
column-field	640	23.8	11.9	0.00
alternating-beams	640	25.1	16.4	3.29
two-support-bridge	640	25.0	17.3	2.00
enclosed-shell	640	25.9	9.0	1.00

Table 4. Full-release geometric means. Multi-sup. counts multiply supported parts per object, not a physical-stability score.

copy-controls.jsonl.gz; selection and artifact hashes are recorded in audit.json. Summary code rejects duplicate cases or repeated sources within an endpoint and records metric-specific denominators. These are descriptive census values on the public release, not sampling intervals or held-out model performance.

2. Additional Measured V2 Results

2.1. Coordinate-Origin Revision

The original 24-call nested 2/4/8-piece ladder retained minimum $Z = 1$ for two prefixes despite inherited minimum-corner wording. One failed response matched up to translation. Original outputs and scores are preserved. A separate zero-anchored, re-rendered 24-call revision has RGB+BOM success 1/12 and symbolic success 11/12. Both ladders derive from one source, not six independent scenes. Origin, wording, image rendering, and re-sampling change together, so the difference cannot be assigned solely to ori-

ID	Catalog type	Instances	Objects
2456	Brick 2x6	0	0
3001	Brick 2x4	1,799	948
3002	Brick 2x3	1,845	963
3003	Brick 2x2	3,859	1,573
3004	Brick 1x2	11,736	2,889
3005	Brick 1x1	28,210	3,540
3007	Brick 2x8	0	0
3008	Brick 1x8	4,805	1,250
3009	Brick 1x6	2,568	611
3010	Brick 1x4	4,865	1,581
3020	Plate 2x4	1,836	584
3021	Plate 2x3	1,991	595
3022	Plate 2x2	6,162	2,166
3023	Plate 1x2	7,221	2,209
3024	Plate 1x1	20,957	4,153
3031	Plate 4x4	0	0
3032	Plate 4x6	0	0
3034	Plate 2x8	0	0
3035	Plate 4x8	5,760	5,120
3460	Plate 1x8	3,528	1,247
3622	Brick 1x3	2,151	985
3623	Plate 1x3	2,217	605
3666	Plate 1x6	2,645	617
3710	Plate 1x4	4,915	1,222
3795	Plate 2x6	0	0

Table 5. Exact catalog coverage in the 5,120 source assemblies. Instances count placements; objects count sources containing a type. Zero-frequency types remain in the 25-type recognition catalog.

gin normalization.

Task / variant	GPT-4.1 mini	Gemini Flash
parts	8/25	12/25
relations/contact	7/24	7/24
relations/separated	0/24	0/24
reconstruct/full	0/24	0/24
generate/constraints	0/24	0/24
complete/suffix	0/24	0/24
edit/rotate	0/24	0/24
edit/recolor	24/24	24/24
edit/remove	24/24	24/24
plan/assemble	1/24	12/24
plan/disassemble	1/24	7/24
repair/none	21/24	17/24
repair/color	4/24	6/24
repair/shift	0/24	0/24

Table 6. The earlier 24-source v2 study, retained separately from the expanded three-source matrix. Recognition uses 25 catalog types. All structural variants share the same 24 source groups.

Copy-current task	Cases	Success	Part F1	Local
complete/suffix	5,120	0.0	87.1	0.0
edit/rotate	5,120	0.0	0.6	0.0
edit/recolor	5,120	0.0	79.0	0.0
edit/remove	5,120	0.0	97.4	0.0
repair/none	5,120	100.0	100.0	100.0
repair/color	5,120	0.0	94.9	0.0
repair/shift	5,120	0.0	94.9	0.0

Table 7. Full-casebank no-operation census. Local is addition F1, edit-target success, or repair-localization F1, respectively. Rows share 5,120 sources; this is not a learned-model test split.

Direction	Split	Cases	Successes	%
assemble	train	2,690	1,615	60.0
assemble	validation	392	230	58.7
assemble	test_id	758	468	61.7
assemble	test_ood	1,280	51	4.0
disassemble	train	2,690	2,690	100.0
disassemble	validation	392	392	100.0
disassemble	test_id	758	758	100.0
disassemble	test_ood	1,280	1,280	100.0

Table 8. Public-input greedy planning, original submitted order. Height sorting separately solves all 10,240 directional cases. Neither algorithm is a learned model.

3. Earlier Neural Pilots

Two preceding releases provide neural pipeline evidence but not a v2 leaderboard. A 576-object composition pilot evaluated four held-out objects with GPT-4.1 mini and Gemini 2.5 Flash. Each model answered 56 cases, for 112 API requests. Table 12 shows visual task counts; all

Gemini control	Original	Control
no-image	0/8	0/8
mismatched-images	0/8	0/8
symbolic-reference	0/8	8/8
reflection	0/8	0/8
validation	0/8	0/8
repeat	0/8	0/8
relative	6/8	0/8

Table 9. Eight-source paired controls. Relative is assembly representation; the remaining arms are reconstruction. The symbolic arm adds privileged geometry.

Condition	Seed	Steps	Answer tokens	Success
base	17	0	0	0/337
leave-edit	17	120	61,440	7/337
leave-edit	29	120	61,440	8/337
leave-edit	43	120	61,440	9/337
multi	17	120	61,440	9/337
multi	29	120	61,440	2/337
multi	43	120	61,440	9/337
single	17	120	61,440	0/337
single	29	120	61,440	0/337
single	43	120	61,440	0/337

Table 10. All ten measured local v2 VLM jobs, 337 cases each. The nine trained runs match supervised answer tokens, not total image/input tokens, compute, or elapsed time.

Condition	Mini	GPT-4.1	Flash	Qwen
Full structure (RGB)	2/4	4/4	2/4	0/4
Local pose (RGB)	3/4	2/4	2/4	1/4
Choice (RGB)	2/4	2/4	2/4	4/4
Choice (permuted)	2/4	2/4	2/4	2/4
Choice (no image)	2/4	1/4	1/4	1/4
Local pose (symbolic)	4/4	4/4	4/4	4/4

Table 11. Measured four-source adaptive probes. Candidate lists, local-pose outputs, and symbolic references are different protocols, not interchangeable scores.

cells have only four objects. Both models failed ordinary and layer reconstruction/completion, but succeeded with the privileged symbolic reference. This suggests a useful perception-versus-structure diagnostic, not a statistically established model comparison. V2 also changes some scoring rules, including full-structure requirements under layer disclosure; cross-version scores must not be pooled.

A further Gemini control reuses the identical first answer and input for 12 ordinary-condition generation/reconstruction/repair cases. One additional call either requests reflection without new scene information or supplies target-blind legality feedback. Both arms remain at

Task	Model	RGB	Layers	Symbolic
reconstruct	GPT-4.1 mini	0/4	0/4	4/4
reconstruct	Gemini Flash	0/4	0/4	4/4
complete	GPT-4.1 mini	0/4	0/4	4/4
complete	Gemini Flash	0/4	0/4	4/4
repair	GPT-4.1 mini	0/4	1/4	3/4
repair	Gemini Flash	2/4	0/4	3/4

Table 12. Earlier composition-v1 pilot only. Gemini Flash denotes Gemini 2.5 Flash. These 112 API calls are not v2 model evaluation.

Condition	Seed	Steps	Tokens	Success
base	17	0	0	0/8
single	17	120	195,292	2/8
multi	17	120	173,957	1/8
multi	29	120	176,765	1/8
multi	43	120	173,137	2/8

Table 13. Earlier Grid-v1 local text/structure pilot, eight test cases per row. Tokens count processed input plus output training tokens. Base/single have one seed each. This is not v2 or multimodal training.

2/12 exact successes, with no gained or lost successes. The 24 new requests are billed once; reused historical receipts are not charged again. Arms ran sequentially rather than interleaved, and the sample is too small to establish that validation tools are generally ineffective.

An older Grid-v1 text/structure pilot uses Qwen3-0.6B and rank-8 LoRA on query/value projections, with 1,146,880 trainable parameters, AdamW learning rate 2×10^{-4} , and 120 optimizer steps per trained run. The maximum sequence length is 3,072; overlength training examples are excluded rather than truncating answers. Greedy inference allows 1,200 new tokens. Single-task training uses editing; multi-task training uses relations, generation, editing, and planning. Table 13 reports actual outputs. All successes are editing cases. Multi-task seeds average 16.7% with a sample standard deviation of 7.2 percentage points, without evidence of broad transfer.

The single/multi runs are step-matched, not token-matched, and use different supervision diversity. A lower validation NLL is not evidence of execution success. Four adapters, five prediction sets, and all 480 optimizer-step records are bundled. Training used an Apple M4 Pro with 48 GB memory and MPS; local computation consumed real resources despite having no API bill.

4. Evidence Boundaries

The six-work-package research contract separates positioning, data, measurement, model experiments, independent

review, and delivery. Same-machine runtime-free replay is not independent reproduction. Review packets contain no completed human judgments until real reviewers submit version-matched labels; disagreement is not silently adjudicated. No human-designed dataset is treated as imported merely because a paper or connector annotation is publicly accessible. Permission, assets and redistribution rights are separate gates.

5. V2 Training And Verification

All 117,910 strict-decoder oracle answers pass, 5,120 malformed primitive-type examples are rejected, and scores on 118,640 existing baseline/API records agree with their historical values. Independent implementations agree on 138,240 structural metric comparisons, all 10,240 relation queries, and connector counts on 5,120 structures. An additional 107,520 mutation/invariance checks cover missing, duplicate, unknown, and reversed plan actions, no-op edits, faulty localization, IDs, and yaw symmetry. Finite tests are not a proof over arbitrary malformed submissions, and shared catalog assumptions limit algorithmic independence.

The local pilot uses SmolVLM-256M-Instruct revision 7e3e67ed, unquantized. Ninety-six training and 24 disjoint validation objects provide seven task families (672/168 examples) and actual PNG views. Language-side query/value rank-8 LoRA has 460,800 trainable parameters. Single-reconstruction, multi-task and leave-edit use seeds 17, 29 and 43. Every trained run uses 120 updates, exactly 512 supervised answer tokens per update, AdamW at 2×10^{-4} , and gradient clipping at 1. Full prompts and answers are retained; deterministic answer-token subsampling matches supervision, not input/image tokens, time or compute.

Validation NLL uses two fixed examples per task. Training uses PyTorch/MPS; formal inference uses MLX with a 2,200-token cap on the same 337 cases. Token, pixel and mask digests agree across preprocessing backends, but floating-point kernels need not produce identical outputs. All 840 training/validation prompt prefixes retain token identity when answers are appended; maximum complete length is 5,176 tokens. Compact test-oracle answers use at most 1,793 tokens. Oracle lengths are private diagnostics, never appended to model inputs. The completed matrix retains 3,370 predictions, 1,080 updates and 552,960 supervised tokens. Each of nine adapters has 60 adapted matrices and 411 unchanged tensors independently checked under the actual bfloat16 cast-before-add merge semantics.

The calibration pool contains 468 questions from 36 validation objects excluded from the original study and training/validation bundle. Selection cycles through available size bins within each of six policies; missing bins remain explicit. Portable inputs contain 720 image uses and 144 PNGs checked against public input, hashes and non-

blank rendering. The earlier train/validation/test render audit checks 1,177 records, 1,970 image uses and 626 regenerated images on the same browser/GPU stack, not arbitrary rasterizers.

The casebank verifier checks 476,760 serialized rows across 14 shards, regenerates every ground truth and reconstructs baseline state chains. Browser checks cover desktop and mobile, nonblank scenes, camera interaction, input/truth isolation and export. Additional development responses are replayed offline and are not retroactively included in older browser-audit counts. Reproduction shares pinned Node dependencies; it is not an independently installed operating system or a second person’s training experiment.

6. Candidate Controls

For the initial pose probes, all candidate structures are legal, use identical part identities, yaw and height, and differ in horizontal position. Correct labels are balanced across the four selected scenes. Repeating a question and permuting its candidates are distinct interventions. The follow-up contract compares permutation mismatch to same-order repeat mismatch within each source/model, with temporal order balanced across source groups. Source-level bootstrap intervals describe only this selected development sample; they do not establish population coverage or stable model rankings.

6.1. Renderer-Aware Analysis by Synthesis

To test whether candidate locations contain usable image signal, a post-hoc deterministic baseline consumes only public BOM, candidate poses and RGB views. For the two-piece blue-on-gray domain, the supplied positive query height and support rules imply a grounded gray piece. Its horizontal positions and both distinct yaw footprints are enumerated under minimum-corner and contact constraints, not obtained from ground truth. Every legal candidate scene is rendered with the public renderer.

We threshold blue chroma ($B > 80$, $B > 1.25R$, $B > 1.1G$) inside image rows 32–447, excluding title/footer and neutral grid/text. A candidate receives the smallest mean mask-IoU distance over its legal hypotheses in top, front and side views. Equal candidate scores or missing blue masks cause abstention. No-image inputs abstain without rendering. The thresholds and enumeration are fixed in code before this run, but the method was designed after the preceding development experiments.

All 44 image-conditioned responses succeed, representing only 16 original sources with repeated/permutated conditions. All 16 permutations preserve the selected pose. Cyclic replacement with the next source’s views causes one abstention and changes 9/15 jointly answered selections; these mismatched BOM/image inputs test dependence on

Development condition	Success	Abstain	Sources
Pose/choice-rgb	4/4	0	4
Pose/choice-permuted	4/4	0	4
Pose/choice-no-image	0/4	4	4
Order/original	12/12	0	12
Order/repeat	12/12	0	12
Order/permuted	12/12	0	12
Order/no-image	0/12	12	12

Table 14. Specialized known-renderer baseline. The 60 conditions reuse only 16 sources. No-image zeros are deliberate abstentions, not random-guess accuracy. This is not an additional neural model.

pixels, not valid-task accuracy. The implementation has no calibrated out-of-domain rejection threshold for nonblank inputs. The run generates 384 candidate views in 29.7 seconds on the local browser stack, including 60 predictions and 16 image replacements, with no API calls. Cached templates enable offline rescoring.

Using the exact renderer, canonical colors and candidate information strongly favors this method. It is an algorithmic simulator-aware control, not a general reconstruction system, novel learned model, independent human validation or confirmation on new sources. Original VLM prompts also contained an isometric view, which this baseline does not use. Prediction production precedes loading evaluator targets for scoring.

7. Resource Accounting

The earlier 363-call campaign cost \$0.45102511, and the 730-call v2 study cost \$1.50635072. The four-model screen and both ladder versions add 228 calls and \$0.695073176; pose probes add 96 calls and \$0.162861212. The 192-call repeat/permutation study costs \$0.282100356. The full campaign totals 1,609 settled calls and \$3.097410574 under the original \$4.50 cap. No ledger is reset between studies. Local training and inference consume compute but do not incur API charges. These are historical campaign totals before the four-model paper expansion. The additional 156 requests and updated cumulative spend are generated from settled receipts in `expanded-evidence.json`; they are not included retroactively in the older campaign tables. The four new models cost \$0.523093880 across 156 settled requests. The cumulative campaign is 1,765 calls and \$3.620504454, below the unchanged \$4.50 cap. The new model allocation was capped at \$0.75; geometry audits and paper renders made no model calls.

8. Expanded Measured Model Evidence

The eight-model main matrix uses the identical 39 tasks from three already-observed development objects. Original GPT-4.1 mini, GPT-4.1, Gemini 2.5 Flash, and Qwen3-VL-

32B responses are reused without rebilling. Claude Haiku 4.5, Gemini 3 Flash preview, Qwen3-VL-8B, and Llama 4 Maverick add 156 requests. Prompts and image bytes are hash-matched across rounds. New source observations are not claimed: these three objects already appear in the cumulative exposure register. Model endpoint IDs, catalog prices, provider routing, raw replies, latencies, tokens, and settled receipts are stored in `study/paper-models/`.

The same-source algorithm rows are generated from public inputs only. Copying applies to current-structure tasks; retrieval searches only training sources using public BOM or extent requirements without target-aware adaptation; symbolic rules solve relations/edits; greedy and height-sorted planners receive only the supplied target. Their supported-task scopes are explicit, and unsupported fields are not reported as failed attempts.

9. Finite-Grammar Ambiguity Contract

The new extension is versioned separately from v2. A fixed shell encloses a $2 \times W$ unknown layer tiled by identical red 1-by-2 bricks. All other layers and blue query poses are fixed. Widths 2, 4, and 6 and three stack heights yield nine parameterized families, not independently collected semantic objects. Every possible query pair is retained, not selected according to model outcomes. The two conditions share exactly the same query and family: the exterior condition withholds the internal layer, while support disclosure reveals its complete placement list.

`public.jsonl` supplies the grammar, known geometry, inventory, query IDs, and (only in the disclosed condition) support poses. It excludes expected labels, answer sets, and alternative hidden programs. Known geometry is symbolic, so this is a test of observation-consistent structural reasoning, not a claim of visual recognition. The public-input solver enumerates legal hidden tilings, forms all candidate contact graphs, and returns a definite answer only if every candidate agrees. It receives no evaluator target.

The interior enumeration uses first-empty-cell backtracking with horizontal and vertical domino placements. Independent recurrence counts validate enumeration completeness for the finite grammar. Every candidate passes both geometry implementations. Independent contact-edge intersections validate shared-support labels. The exact answer set is not claimed exhaustive over arbitrary brick programs or non-grid connectors.

False certainty counts definite yes/no on an undetermined input; unnecessary abstention counts undetermined on a determinate input. Malformed responses have zero exact success and separate format failure. Neither conditional rate has a denominator on inapplicable cases. Because answer classes are unbalanced, overall accuracy, source-macro accuracy, and class-balanced accuracy must all be reported.

Pixel audit. All 60 layout states are rendered in four fixed views with the separate high-resolution paper renderer, for 240 frames. Thirty-six serve as the first-layout reference of their family/view, leaving 204 alternative comparisons. All alternative pixels match, with no tolerance. This does not certify equality under a different renderer, transparent parts, changed cameras, or actual photography. It also does not establish human agreement on the grammar or instructions.

Figure integrity. Thirty actual 3D renders form four new figure plates. Gallery examples are selected by median part count and deterministic group hash. Task examples preserve original geometry, colors, and evaluator IDs; all oracle targets are scored before rendering. Execution snapshots are verified legal prefixes, while the greedy dead end is checked against the original insertion rule. Fixed framing prevents camera rescaling from masquerading as a change in part pose. Exploded layers are explicitly display-only and are never substituted for ordinary model observations. Non-blank pixels, uncut framing, render recipes, and file hashes are recorded in `structure-evidence.json`.

Variant	Mini	GPT-4.1	Flash 2.5	Qwen 32	Haiku	Flash 3	Qwen 8	Llama 4
relations/contact	3/3	1/3	2/3	0/3	0/3	2/3	1/3	0/3
relations/separated	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3
reconstruct/full	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
generate/constraints	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3
complete/suffix	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
edit/rotate	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3
edit/recolor	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3
edit/remove	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3
plan/assemble	1/3	2/3	1/3	0/3	0/3	2/3	0/3	1/3
plan/disassemble	1/3	0/3	1/3	0/3	0/3	2/3	0/3	0/3
repair/none	3/3	1/3	2/3	1/3	2/3	0/3	1/3	0/3
repair/color	1/3	1/3	1/3	0/3	0/3	1/3	0/3	1/3
repair/shift	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3

Table 15. All measured task variants in the expanded screen. Each cell is successes out of the same three objects. No uncertainty interval or reliable ranking is inferred from three clusters.

Condition	Method	Correct / cases	Balanced acc.	Source-macro acc.
Exterior	Set-valued oracle	300/300	100.0	100.0
Exterior	Public-input enumeration	300/300	100.0	100.0
Exterior	Always no	210/300	50.0	57.8
Exterior	Always undetermined	90/300	50.0	42.2
Exterior	Single-witness answer	210/300	50.0	57.8
Disclosed	Set-valued oracle	300/300	100.0	100.0
Disclosed	Public-input enumeration	300/300	100.0	100.0
Disclosed	Always no	264/300	50.0	81.1
Disclosed	Always undetermined	0/300	0.0	0.0
Disclosed	Single-witness answer	300/300	100.0	100.0

Table 16. Measured algorithmic controls for the new ambiguity extension, percentages. Balanced accuracy averages only the answer classes present in that condition. The single-witness baseline commits to one valid layout even when the observation admits alternatives.

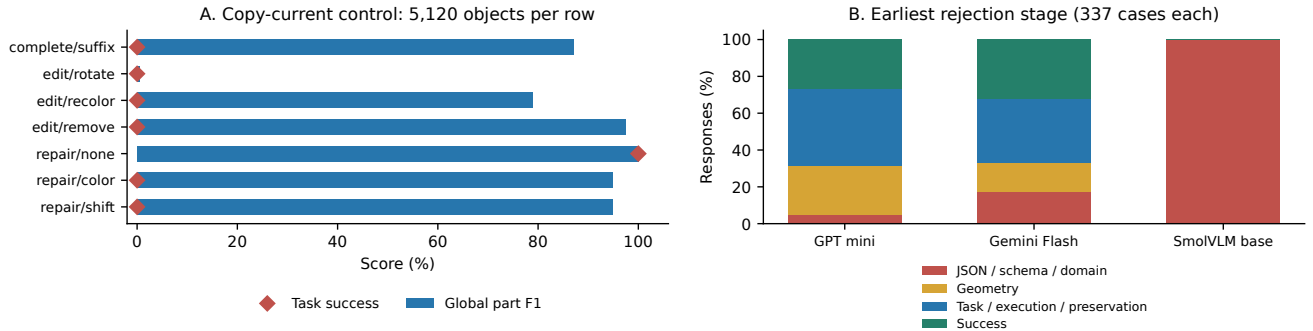


Figure 6. Previously measured no-operation and failure-stage diagnostics. The left panel uses the full casebank; the right uses the 337-case two-model/local-base study, not the new three-source matrix.

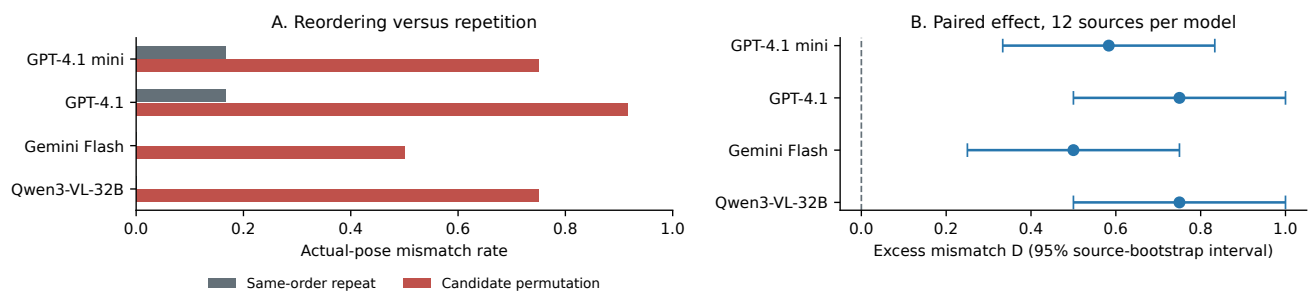


Figure 7. Measured same-order repetition versus candidate permutation on 12 shared development sources. Intervals resample source groups and do not constitute independent population confirmation.

10. Prospective Experiments: No Results Yet

This section is a work plan. Every result cell below is deliberately blank, and its machine-readable value in `experiments.json` is null. Tables are generated with an assertion that no prospective cell contains a measured number. They are not an extension of the completed pilots, a preregistration, or evidence of dataset acquisition. Tasks and models need final revision hashes and source manifests before execution. A model name here indicates an intended comparison, not confirmation that a future provider will serve an identical checkpoint.

10.1. Independent Main Evaluation

Table 17 gives the independent-confirmation layout. All methods must share source groups for paired comparison. Report ordinary RGB+BOM reconstruction/completion/repair separately from symbolic relations/editing/planning and constraint generation. Catalog recognition uses a separate type-cluster denominator. Each eventual score requires actual sources, failures, estimate, and 95% source-level interval. No pooled eight-task rank is prespecified. Sample size follows a predeclared precision target. For illustration, Hoeffding’s bound with a union bound over four means of $D_g \in [-1, 1]$ gives $n \geq \lceil 2 \log(8/\alpha)/\epsilon^2 \rceil$ independent sources for simultaneous absolute error at most ϵ with probability at least $1 - \alpha$. At $\alpha = .05, \epsilon = .10$, this is 1,016 sources and 16,256 calls for four models with four arms each. This is a conservative planning bound, not an acquired dataset, a guaranteed future effect, or a universal required sample size. Stratified/variance-based designs require their own frozen assumptions.

Method / condition	N	Part	Rel.	Recon.	Gen.	Compl.	Edit	Plan	Repair
GPT-4.1 mini									
GPT-4.1									
Gemini 2.5 Flash									
Qwen3-VL-32B									
Claude Haiku 4.5									
Gemini 3 Flash (preview)									
Qwen3-VL-8B									
Llama 4 Maverick									
Additional open VLM									

Table 17. **Unmeasured.** Independent eight-task confirmation. Blank cells are unrun results, not zero scores. N is the actual source denominator, with recognition reported separately by catalog type. Each cell requires an estimate and source-level interval; task variants and input conditions must also be broken out in the result artifacts.

10.2. Information and Representation

Table 18 is repeated per model on the same sources. Report the ordinary surface criterion for all arms, but full hidden reference success only with its information condition explicitly stated. Local pose narrows the requested output; candidates additionally restrict the answer space. They cannot isolate output length without that confound. For feedback versus reflection, reuse the same first response and charge the additional request once. Feedback remains target-blind; matching call counts alone does not match compute.

Method / condition	N	Success	Part F1	Surface F1	Format	Tokens
RGB+BOM / full structure						
BOM only / full structure						
Layer disclosure / structure						
Symbolic reference / structure						
RGB+BOM / local pose						
RGB+BOM / relative program						
RGB+BOM / legality feedback						
RGB+BOM / reflection only						

Table 18. **Unmeasured.** Paired information/representation ablations, separately per model. Cells need measured values and applicable denominators; blank is not zero.

10.3. Candidate Robustness and Debiasing

Table 19 compares original, repeated, and permuted actual pose selections on independent sources. Correct label balance and request ordering must be frozen. Missing responses reduce paired coverage, not the denominator of service availability. Report complete-pair effects alongside worst-case bounds: with N frozen sources, n complete pairs, and observed effect sum s , the all-source effect lies in $[(s - (N - n))/N, (s + (N - n))/N]$. This is a missing-outcome bound, not a confidence interval. For uniform independent random K -way choice, accuracy is $1/K$; repeat and permutation mismatch expectations are both $1 - 1/K$, so expected excess mismatch is zero. These are analytic controls, not empirical model results.

An option-debiasing baseline is motivated by existing selection-bias work and must be fitted on development data only. API conditions that do not expose token probabilities require a declared compatible alternative; one must not silently substitute a different method. Simulator-aware matching receives renderer access and should be reported with that additional resource.

Method / condition	Pairs	Accuracy	Repeat	Permute	D	95% CI
GPT-4.1 mini						
GPT-4.1						
Gemini 2.5 Flash						
Qwen3-VL-32B						
Dev-fitted option debiasing						
Known-renderer matching						

Table 19. **Unmeasured.** Independent candidate robustness. Repeat and Permute are pose-mismatch rates relative to the original, not error rates against ground truth; D is their paired difference.

10.4. Difficulty and Nuisance Factors

Table 20 must be instantiated per task/model. Cross piece count with policy, visibility, and graph complexity; do not call a marginal size trend causal. Nested prefixes stay in one source cluster. Render perturbations preserve geometry and receive new render/version hashes. Occlusion must be operationalized before selection (for example, visible cells or human-rated visibility), not assigned after observing model errors. Domain-specific local metrics include added-part F1, faulty-ID F1, or per-axis pose accuracy.

Method / condition	Sources	Success	Local F1	Format	95% CI
2–8 pieces / calibrated					
9–16 pieces / matched policy					
17–28 pieces / matched policy					
29–47 pieces / matched policy					
Canonical camera and palette					
Camera perturbation					
Palette perturbation					
Occlusion-matched structures					

Table 20. **Unmeasured.** Difficulty and camera/palette controls. Missing policy-size strata remain missing; they are not generated after seeing a desirable model ranking.

10.5. External Sources and Specialized Methods

Table 21 is contingent on actual licensed assets. Record source revision, file hashes, family IDs, permitted uses, unsupported part rejection, coordinate and connector mapping, and redistribution scope. A code license or a public paper does not grant rights to a separate structure collection. Do not flatten unsupported parts into cuboids and claim full-domain coverage.

MEPNet requires stepwise manual and component inputs, BrickGPT a compatible text-generation target, and BrickNet connector programs. Scores are meaningful only after task and input adaptation is audited. Native metrics and BrickAtlas metrics must be separately named. Unsupported cells should eventually be marked not applicable, not zero. No published score from another dataset may populate this table.

Method / condition	Sources	Types	Import rej.	Success	95% CI
Licensed human designs / VLM					
Licensed human designs / retrieval					
Manual-step subset / MEPNet adapter					
Text-generation subset / BrickGPT					
Connector subset / BrickNet					

Table 21. **Unmeasured.** External-domain and specialized methods. Import rej. is the rejected-source proportion with explicit reasons. No external structure collection or adapter run is claimed complete.

10.6. Human Metric Calibration

Table 22 requires at least two real independent, blinded reviewers and documented third-person adjudication when needed. Reviewers inspect public observations, candidate output, and the reference without automatic scores. An alternative answer may be correct despite differing from the reference. Preserve uncertain judgments and all original disagreements.

False acceptance uses human-rejected outputs as denominator; false rejection uses human-accepted outputs. Report both counts, source-cluster intervals, agreement before adjudication, unresolved coverage, and task strata. Missing denominators remain unestimable, not zero. Software verifies packet/version hashes, not real reviewer identity or independence. Current review packets cover 96 candidates from 48 cases and eight sources, with zero submitted labels; extending the packet is not equivalent to conducting the review.

Method / condition	Sources	Outputs	False acc.	False rej.	Agreement
Reference-correct outputs					
Alternative hidden layouts					
Near-correct geometry					
No-op edits and completions					
Normal and faulty repairs					
Executable and blocked plans					
Licensed external structures					

Table 22. **Unmeasured.** Human calibration categories. Outputs and independent sources are different denominators. No simulated or model-authored labels count as human evidence.

10.7. Effective Training and Replication

Table 23 describes a new study, not missing measurements from the ten completed jobs. First establish that the chosen checkpoint can use the interface and achieve nontrivial task performance. Freeze source-disjoint training/validation/test sets, model revision, supervised answer tokens, optimizer steps, and seeds. Also report image/input tokens, memory, and elapsed compute. Leave-task-out transfer needs an actual held-out-task improvement; lower validation loss alone is insufficient.

Independent reproduction uses a second executor and separately installed dependencies. Record commit, OS/architecture, versions, commands, exit codes, output hashes, and discrepancies. Separate offline rescoring, re-rendering, retraining, and new API replication: none certifies the others. Existing cumulative paid-call budgets remain in force.

Method / condition	Seeds	Success	Held-task gain	95% CI	Compute
Effective base checkpoint					
Single-task adaptation					
Multi-task adaptation					
Leave-edit-out adaptation					
Second-environment replication					

Table 23. **Unmeasured.** Effective training and independent replication. Held-task gain must identify its reference condition and paired source population.

10.8. Equal-budget tools and correction

Freeze one shared first response for each case. Distinguish a new model request from a replayed response. Compare equal call budgets and report actual token/time/renderer budgets. Only target-blind feedback is allowed; do not supply the hidden reference as tool feedback.

Method / condition	N	Success	Local F1	Legal prefix	Cost
One-shot / no tool					
One extra reflection					
One legality-check response					
Render own prediction + revise					
Active additional view					
Execution feedback + replan					
Fixed three-call tool budget					

Table 24. **Unmeasured.** Equal-budget tools and correction. Blank cells are unrun results, not zeros.

10.9. Representation and decoder factorial

Use identical source objects and observation payloads. Cross output representation with decoder support rather than changing them simultaneously. Relative offsets in global axes differ from local connector coordinates. A graph-backed adapter requires verified connector semantics; unsupported parts must be reported.

Method / condition	N	Format	Pose F1	Edge F1	Success
Absolute JSON / free decoding					
Absolute JSON / schema decoding					
Global-relative / free decoding					
Global-relative / schema decoding					
Connector graph / audited adapter					
Incremental placement / same images					

Table 25. **Unmeasured.** Representation and decoder factorial. Blank cells are unrun results, not zeros.

10.10. Ambiguity-aware model evaluation

New finite-grammar extension only, not v2. Evaluate exterior and support-disclosed arms separately with matched source families. Report class-balanced success because no/yes/undetermined are unequal. False certainty is conditional on ambiguous inputs; unnecessary abstention is conditional on determinate inputs. Do not equate nine grammar families with independent semantic sources.

Method / condition	Families	Balanced acc.	False certainty	Abstention	Disclosure gain
GPT-4.1 mini					
GPT-4.1					
Gemini 2.5 Flash					
Gemini 3 Flash (preview)					
Claude Haiku 4.5					
Qwen3-VL-8B / 32B					
Llama 4 Maverick					
Explicit candidate-layout enumeration					

Table 26. **Unmeasured.** Ambiguity-aware model evaluation. Blank cells are unrun results, not zeros.

10.11. Planning scope and harder action domains

Current vertical full-target planning is solved by height sorting; keep that as a ceiling control. New action domains need separately versioned geometry and independently verified witnesses. Do not add an obstacle or nonvertical action to the old evaluator and compare scores as if the task was unchanged.

Method / condition	Sources	Actions	Success	Search nodes	Time
Vertical full-target / neural baseline					
Partial-target / dependency inference					
Removal for access / repair sequence					
Workspace obstacles / search					
Typed connector / articulated action					

Table 27. **Unmeasured.** Planning scope and harder action domains. Blank cells are unrun results, not zeros.

10.12. Task-adapted comparisons and cross-benchmark validity

These are unimplemented or unrun comparisons, not published method scores. Obtain actual licensed assets and pin native evaluator revisions. Report native metrics and BrickAtlas metrics separately, with supported input/task mappings, rejection rates, and training overlap. Do not produce a pooled rank across incompatible benchmarks.

Method / condition	Sources	Native metric	Adapted success	Reject rate	Compute
BC-Bench / Brick-Composer					
MEPNet / manual-step adapter					
Neural Assembler / multi-view					
TreeSBA / multi-view adapter					
BrECS / budgeted completion					
BrickGPT / semantic generation					
BrickNet / connector generation					
SpatialBabel / scene-code control					

Table 28. **Unmeasured.** Task-adapted comparisons and cross-benchmark validity. Blank cells are unrun results, not zeros.

11. Artifact and Figure Reproduction

From the repository root, run the following with Node 22 and the documented Python figure dependencies:

```
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/study/paper-audit.ts
node benchmark/paper/generate-tables.mjs
node benchmark/paper/generate-study.mjs
node benchmark/paper/generate-audit.mjs
python benchmark/paper/generate-figures.py
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/study/ambiguity.ts --verify
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/study/paper-models.ts replay
node node_modules/tsx/dist/cli.mjs \
  benchmark/paper/generate-expanded.ts
node node_modules/tsx/dist/cli.mjs \
  benchmark/paper/render-structures.ts
python benchmark/paper/compose-structures.py
node node_modules/tsx/dist/cli.mjs \
  benchmark/suite/challenge/render.ts
python benchmark/suite/challenge/compose.py
node node_modules/tsx/dist/cli.mjs \
  benchmark/paper/generate-challenge.ts
```

Then compile both TeX documents using the instructions in `paper/README.md`. `audit-evidence.json` binds new tables to the census and prospective plan; `figure-evidence.json` binds plots to source files and output PDFs. The copy-control case-level file supports split/policy rescoring. The figure generator uses real archived PNGs, vector plots of observed data, and an explicitly disclosed cross-section of a verified witness.

The new audit does not modify frozen dataset generation, task labels, old responses, or legacy scorer behavior. It adds measurement coverage, not additional independent sources. Successful compilation, hash checks, and source-bound figures are technical release checks; they do not clear the independent scientific-validation gates. New three-dimensional plates use a separate renderer so historical evaluation images and their hashes remain unchanged. The optional render command needs the existing benchmark development server; all other commands above are offline after dependencies are installed.